Automated Pothole Detection and Bounding-Box Annotation Using Deep Learning

Proposed methodology and comparative study | October 2026

Abstract

This project proposes an accurate and efficient system for automatically annotating potholes in approximately 209 roadside-perspective images. The required output is an axis-aligned rectangular bounding box around each pothole, not a pixel-wise segmentation mask. Approaches considered include training a detector from scratch, fine-tuning general-purpose detectors, using pothole-specific pretrained models, and adapting SAM 3 with LoRA. **The preferred approach is a pothole-pretrained RF-DETR model, followed by targeted fine-tuning and human verification where needed.** SAM 3 is retained as a comparative research direction.

1. Problem Statement

Given a roadside image, the system should predict zero or more pothole bounding boxes, each with coordinates (x_min, y_min, x_max, y_max) and a confidence score. The system should minimize missed potholes, false detections, and inaccurate box localization.

Key challenges: Perspective distortion; irregular pothole shapes; confusion with shadows, cracks, water, patches and worn asphalt; large variation in apparent pothole size; and a small target dataset.

2. Candidate Approaches

Approach	Advantages	Limitations / assessment
Training from scratch	Architectural flexibility	Data-hungry; high overfitting risk. Not recommended.
Fine-tune YOLO	Simple training and fast inference	Useful baseline; may struggle with difficult small potholes.
Pothole-pretrained RF-DETR	Relevant prior training; straightforward deployment	Potential domain shift. Recommended first.
Fine-tune D-FINE	Localization-focused detection architecture	Additional training/configuration; strong alternative.
SAM 3 zero-shot	Text-conditioned detection without labels	Inconsistent confidence and possible false positives.
SAM 3 + LoRA	Parameter-efficient domain adaptation	Higher complexity and GPU requirements; advanced experiment.

3. Relevant Research and Datasets

HRP4K (2026): A high-resolution perspective-view road image dataset containing 6,003 images and 7,217 annotated potholes. Its acquisition viewpoint and bounding-box task make it a relevant source of pretrained models and additional training data. The published benchmark reported the following performance on HRP4K:

Detector	mAP@50	mAP@50-95
YOLOv5	55.5%	34.3%

Detector	mAP@50	mAP@50-95
YOLOv8	58.0%	39.0%
YOLO11	59.6%	40.7%
RT-DETRv1	60.6%	37.0%
RT-DETRv2	60.5%	37.3%
D-FINE	61.1%	38.3%

These results are published benchmark measurements, **not** results on the target 209-image dataset. D-FINE leads in mAP@50, whereas YOLO11 leads in mAP@50-95. No single detector dominates every localization metric.

4. Proposed Methodology

Phase 1 - Dataset preparation. Inspect image orientation, quality, resolution, duplicate scenes, lighting, pothole scale and negative examples. Preserve useful road texture and avoid aggressive preprocessing.

Phase 2 - Pretrained comparison. Evaluate HRP4K-pretrained RF-DETR Small and zero-shot SAM 3 on a representative manually annotated subset. Use the same images and annotation rules for a fair comparison.

Phase 3 - Selective fine-tuning. If accuracy is inadequate, fine-tune the stronger model on a separate, human-corrected training subset. Use moderate augmentation, pretrained weights, early stopping and validation-based checkpoint selection. Consider LoRA for SAM 3.

Phase 4 - Resolution experiments. Compare standard full-image inference with overlapping tiles, remapping detections to original coordinates and suppressing duplicate boxes. This may improve recall on small distant potholes at the cost of additional compute and potential false positives.

Phase 5 - Human-in-the-loop annotation. Run the selected detector over the remaining images, correct missing and inaccurate boxes, and export annotated images plus YOLO or COCO annotation files. Optionally repeat fine-tuning using verified labels.

5. Evaluation Plan

Metric	Purpose
Precision	Fraction of predicted potholes that are correct.
Recall	Fraction of actual potholes successfully detected.
mAP@50	Detection quality at IoU threshold 0.50.
mAP@50-95	Detection/localization quality across stricter IoU thresholds.
False positives per image	Annotation cleanup burden.
Inference time	Practical throughput for batch annotation.

Intersection over Union (IoU) is the area of overlap between predicted and ground-truth boxes divided by their union. Prioritize recall to reduce missed potholes, and mAP@50-95 to measure box quality. Hold out a test set that is never used for training or model selection; group images by road/location/session where possible to avoid leakage.

6. Recommended Approach

Primary recommendation: HRP4K-pretrained RF-DETR Small with selective fine-tuning. It can be tested immediately, leverages pothole-specific visual representations, and provides a practical training/inference pipeline for a limited dataset. Start with the public checkpoint, measure performance on manually labeled target images, and fine-tune only if needed.

Research alternative: SAM 3 + LoRA. SAM 3 can output rectangular detections independently of its segmentation masks. Its adaptation is worth investigating if pretrained RF-DETR misses visually ambiguous potholes, but published performance on another dataset does not establish superiority here.

7. Expected Outcomes and Limitations

The proposed system should deliver images with visible pothole boxes, confidence scores, and reusable annotation files. It should reduce manual labeling effort and permit iterative improvement through corrected labels. Likely limitations include domain shift, very small distant potholes, confusing road textures, and imperfect annotation consistency. No accuracy on the target dataset is claimed before evaluation.

8. Conclusion

For a small collection of roadside-perspective images, building a new object detector is unlikely to be the most effective route. A pothole-pretrained RF-DETR checkpoint, evaluated carefully and fine-tuned selectively, offers the strongest practical starting point. SAM 3 and LoRA provide a useful second research track. Human verification remains essential for producing reliable final annotations.

References

Chen et al. (2026). A High-Resolution Perspective-View Road Image Dataset for Pothole Detection. Scientific Data. https://doi.org/10.1038/s41597-026-07317-w

D-FINE: Redefine Regression Task in DETRs as Fine-grained Distribution Refinement. https://arxiv.org/abs/2410.13842

RF-DETR: Real-Time Detection Transformer. https://arxiv.org/abs/2511.09554

SAM 3 official repository. https://github.com/facebookresearch/sam3

HRP4K RF-DETR Small checkpoint. https://huggingface.co/dronefreak/hrp4k-rfdetr-small